

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-979BBA8A02

Country: Botswana

Submission Date: 2025-08-03

Product: Ritomed (ritonavir) | ATC: J05AE03 | Strength: 100 mg

Applicant: Prime Therapeutics | Manufacturer: Nile Generic Medicines

Policy Label: reject_and_return | Risk Score: 0.92

AMR Stewardship: AWARe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable
| Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

This dossier constitutes a formal application for Marketing Authorization of Ritomed (ritonavir) submitted on 2025-08-03 to the national regulatory authority of Botswana. The applicant, Prime Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 100 mg, classified under ATC code J05AE03. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Regional administrative forms were checked for signature completeness and dossier version alignment. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Administrative cover letters, declarations, and fee records were cross-checked against the submission index. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A comprehensive table of contents and glossary of abbreviations are provided to facilitate regulatory review. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intend

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=partial

Error tags: gmp_outdated

Latest GMP inspection date is 2020-09-11. No recent inspection evidence was provided for the required review window. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Batch manufacturing records and standard operating procedures (SOPs) are maintained and archived on-site. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Site quality-system commitments were compared against the proposed commercial manufacturing activities. The applicant confirms alignment with applicable technical guidance and regional submission standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of

PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using t

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=correct

The proposed proprietary name for this medicinal product is Ritomed, containing the active pharmaceutical ingredient ritonavir. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The applicant confirms alignment with applicable technical guidance and regional submission standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were r

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for ritonavir. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Method validation records and quality controls were reviewed for internal consistency. Where applicable, protocol deviations were assessed for impact on interpretation o

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: clinical_missing

Clinical overview for community acquired pneumonia indicates pivotal efficacy data are not available at submission time. Benefit-risk cannot be concluded pending full clinical study reports and verified endpoint analyses. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk

management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The target population demographics are broadly representative of the intended commercial patient cohort. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=present; length=length_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Batch analysis, validation, and impurity management records support the proposed API quality framework. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were reviewed for internal consistency. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. All referenced documents are available in the dossier annex and traceable by section identifier. The API package links specification limits to validated analytical methods and impurity-control decisions. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. Critical material attributes and control strategy elements were reviewed for consistency across the quality dossier. Statistical significance was set at an

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and

reproducibility of the commercial manufacturing operations. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. All re

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable technical guidance and regional

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Pharmacokinetic parameters were derived using non-comp

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

Tabular listing indicates planned or ongoing studies, but no completed pivotal study reports were provided in this submission package. The applicant confirms alignment with applicable technical guidance and regional submission standards. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. The risk management plan includes routine pharmacovigilance and proposed educational materials

for healthcare professionals. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were reviewed for internal consistency. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Method validation records and quality contr

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=missing; length=missing; correctness=incorrect

Error tags: clinical_missing

[MISSING SECTION]

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Stability testing wa